

MAHMMED UMAR FAARUK

AI and Machine Learning Engineer

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PROFESSIONAL SUMMARY

B.Tech. graduate in Computer Science and Engineering (AI and ML) with hands-on experience building and deploying machine learning models, data pipelines, and cloud-integrated applications on AWS. Published researcher in deep learning-based object detection (93.5% mAP, 60 FPS with YOLOv11) with proven experience in real-time monitoring and automated incident-response pipelines. Skilled in engineering resilient, cloud-synchronised systems for low-connectivity field environments. Proficient in Python, PyTorch, scikit-learn, NumPy, Pandas, FastAPI, and AWS. Seeking to apply production ML engineering skills to supply chain and logistics AI at Kaleris.

TECHNICAL SKILLS

Languages: Python, SQL (basic)

ML and Data Libraries: scikit-learn, NumPy, Pandas, Matplotlib, PyTorch, CNN, LSTM, Federated Learning, Transfer Learning

ML Domains: Supervised and Unsupervised Learning, Classification, Regression, Computer Vision, Object Detection (YOLOv11, OpenCV), NLP (TF-IDF, Tokenisation, Word Embeddings)

Cloud, Data and Deployment: AWS (ap-south-1), FastAPI, REST API design, data pipelines (ingestion, preprocessing, augmentation, evaluation), offline-sync / cloud synchronisation, Render, Streamlit Cloud, Docker (familiar)

Databases: PostgreSQL, Supabase, Google Firestore

Generative AI and LLMs: Prompt Engineering, LangChain, Dify.ai, LLM fine-tuning workflows

Dev Practices: Git, MLOps service-hardening, real-time monitoring and alerting, unit testing fundamentals, technical documentation

EXPERIENCE

AI Developer Intern | Viswam AI (Swecha, IIIT Hyderabad and Meta) May 2025 – Jul 2025

- Architected and deployed a cloud-integrated platform on AWS (ap-south-1) with a secure FastAPI and HTTPX API gateway, ensuring no credentials or user data were exposed client-side — applying production MLOps service-hardening practices relevant to enterprise logistics systems.
- Engineered an offline-sync mechanism with automatic cloud synchronisation to Supabase PostgreSQL, improving platform reliability and accessibility in low-connectivity environments comparable to warehouse, port, and field operations.
- Built and deployed an LLM-powered chatbot by integrating an inference API (Dify.ai) with a Streamlit front-end, managing end-to-end data flow from user input to model response.
- Contributed to India's first Telugu Large Language Model (LLM) initiative, gaining direct exposure to NLP data pipelines, prompt engineering, and LLM fine-tuning workflows.

Web Development Intern | Intrainz Oct 2024 – Dec 2024

- Designed and delivered three functional JavaScript web applications (calculator, e-commerce storefront, task manager), applying DOM manipulation, state management, and mobile-first responsive design.
- Built a personal portfolio site from scratch using semantic HTML, CSS, and JavaScript, demonstrating end-to-end project delivery and technical documentation skills.

PROJECTS

Fire and Smoke Detection System (Published Research)

Tools: PyTorch, YOLOv11, Python, OpenCV, Twilio SMS API

- Designed and trained a real-time computer vision pipeline achieving 93.5% mAP accuracy at 60 FPS, suitable for production safety-monitoring environments such as warehouses and industrial facilities.
- Built a complete data pipeline covering ingestion, preprocessing, augmentation, labelling, and model evaluation; benchmarked YOLOv11 against multiple baselines using mAP, F1-Score, Precision, and Recall.
- Integrated automated SMS incident alerting via Twilio API, delivering a real-time model monitoring and incident-response workflow.
- Research accepted for publication in an AI Healthcare book (2025, In Press).

Craft Connect - Cloud-Integrated Full-Stack Platform

Tools: Python, FastAPI, Supabase, PostgreSQL, AWS, Streamlit, JavaScript

- Architected and deployed a mobile-first platform on AWS with a FastAPI backend and PostgreSQL via Supabase; integrated external APIs for verified data contributions.
- Engineered a custom offline-sync mechanism with automatic cloud synchronisation, optimising for the low-connectivity conditions typical of logistics and field operations.

Adaptive AI-SIEM - Federated Cyber Threat Detection

Tools: PyTorch, CNN, LSTM, Federated Learning, Django, TF-IDF, Blockchain

- Built a multi-model deep learning system (CNN and LSTM) on benchmark datasets NSL-KDD and CICDS2017, achieving 99% accuracy and a 0.01 False Positive Rate.
- Integrated Federated Learning for distributed model training across edge nodes without sharing raw data, demonstrating privacy-preserving ML architecture applicable to multi-site logistics networks.
- Conducted structured comparative evaluation against classical algorithms (SVM, Naive Bayes), producing reproducible experiment benchmarks aligned with MLOps best practices.

Chess Buddy - AI-Powered Web Application

Tools: Python, Streamlit, Stockfish Engine

- Integrated the Stockfish engine into a Python backend for real-time computation with an interactive Streamlit UI managing state, rule enforcement, and outcome detection; deployed on Streamlit Community Cloud.

EDUCATION

B.Tech., Computer Science and Engineering (AI and ML)

Graduated: July 2025

Sreyas Institute of Engineering and Technology, Hyderabad | CGPA: 6.28

CERTIFICATIONS AND PUBLICATIONS

Certifications: AWS APAC Solutions Architecture Virtual Experience - Forage (Jan 2025); Machine Learning Workshop - Innomatics Research Labs (Nov 2025); Generative AI Workshop - Growth School (Mar 2026); Goldman Sachs Software Engineering Virtual Experience - Forage (Jul 2025); Tata Data Visualization Job Simulation - Forage (Jul 2025); Machine Learning Workshop - IEEE Sreyas (Dec 2023).

Publication: Employing Deep Learning Paradigms for Fire and Smoke Detection - Accepted, AI Health Care book, 2025 (In Press).

Technical Articles: NumPy Basics: A Practical Guide to Understanding NumPy with Real-World Case Studies (2026); Mastering Python Functions (2026).